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Polarization, mirrors, and reciprocity: birefringence and its compensation in optical retracing circuits

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Birefringence affects all the optical circuits both in free-space and guided-wave optics. It perturbs the state of polarization of the propagating light, causing unwanted detrimental effects in many practical situations. A retracing circuit offers the potentiality of compensating the birefringence, but it is not universal because the birefringence is of different kinds, that is, either linear or circular, as well as either reciprocal or nonreciprocal, and because the common mirror does not hold all the requested symmetries. This paper reviews the compensation techniques suitable for each kind of birefringence, taking into account the introduction of two further generalized mirrors, the mirrored Faraday rotator and the mirrored quarter-wave plate. The main retracing schemes are analyzed and presented with the help of the Poincaré sphere and the Jones matrices. Examples of full compensation of all the main cases of birefringence that occurs in practical optical circuits are given. The different compensation properties of the three mirrors can be interpreted by means of a unified vision in the abstract space related to the Poincaré sphere representation. © 2017 Optical Society of America

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1. INTRODUCTION

It is well known that plane waves with electric and magnetic fields forming an orthogonal triad of vectors together with the wave vector $\mathbf{k}$ are ideal solutions of the Maxwell's equation in free space. Such plane waves are transverse electromagnetic (TEM) modes, and each free-space light beam can be expressed as a superposition of plane waves. For the quantum theory, we can describe each mode as a collection of elementary quanta, i.e., the photons, with well-defined energy:

$$E = \hbar\omega, \quad (1)$$

where ω is the pulsation of light and $\hbar$ is Planck's constant divided by 2π . During the propagation of a light beam in a homogeneous and isotropic medium, we have the conservation of linear and angular momentum, in addition to the conservation of energy [1,2]. In particular, photons corresponding to a circularly polarized plane wave have well-defined linear momentum $\mathbf{p}$ and component s of the spin angular momentum along the propagation direction, given by

$$\mathbf{p} = \hbar\mathbf{k}, \quad (2)$$

$$s = \hbar\sigma, \quad (3)$$

where σ is the spin quantum number that assumes the values $+1$ and -1 for the right and left circularly polarized waves, respectively. Combining the two opposite circular states, with different complex coefficients (i.e., different amplitudes and phases), it is possible to generate every elliptical state of polarization. Therefore, any state of polarization can be described by a bidimensional complex vector. It is then possible to obtain different values of the two complex scalar components of the polarization vector by choosing different pairs of basis vectors. In the usual Jones representation, the basis states of polarization are the horizontal and vertical states, describing a linearly polarized light with an electrical field oscillating along the Cartesian axes x and y , respectively, which are transverse to the propagation axis z .

The above description of the polarization of light, based on the transverse components of the electric field or equivalently on the photon spin angular momentum with respect to the propagation direction, applies not only to the ideal case of the plane waves, but also to the nearly TEM optical beams given by the paraxial beams (like Laguerre–Gauss beams) and the fiber modes in weak-guidance condition (typical of the optical fibers for communications), where the electrical field is nearly transverse, because the longitudinal field component is small [3]. Such beams are exploited in optical communications for transporting information in a specific direction, so introducing a cylindrical frame. If the propagation medium (as an optical fiber or a series of bulk-optic elements) presents a cylindrical symmetry, that is, an invariance for

rotations around the propagation axis z , then the spin along the axis z is conserved [4,5]. An optical fiber gently laid in a straight line is close to realizing this cylindrical symmetry and shows a small perturbation of the state of polarization during propagation.

What happens when the above symmetry is removed, as for instance when the fiber undergoes bending or asymmetric stress? In this case, a linear birefringence is induced, introducing an anisotropy and losing the conservation of the spin. In terms of transverse fields, the linear birefringence causes a phase shift between two orthogonal linear eigenstates of polarization and, hence, a variation of any state of polarization different from the two above eigenstates. Another kind of birefringence is circular birefringence, induced, for example, by fiber twisting, which does not break the cylindrical symmetry and hence conserves the photon spin angular momentum with respect to the propagation axis, but changes any not-circular state of polarization. In fact, the circular birefringence causes a phase difference between its two circular eigenstates, producing a rotation of the polarization azimuth, but a conservation of the polarization ellipticity (related to the spin along axis z).

The polarization fluctuations due to the presence of time-dependent birefringence in the transmission medium are detrimental for several applications in contemporary optics, because they are transformed in a fading of the optical intensity in the presence of polarization-sensitive elements. In the following we illustrate with three examples of how the awareness of the existence of birefringence compensation circuits is useful in different fields of optics.

An example was recently given in the field of the passive optical networks (PONs). Here, the target was to realize a color-less WDM-PON, thus a WDM-PON sharing the same kind of devices in all the remote receivers/transmitters. The solution proposed was to realize a self-tuning laser, which uses part of the leading optical fibers as a laser cavity. The first realizations showed that the uncontrolled birefringence of the circuit introduced severe drawbacks related to the random change of polarization affecting the recirculating amplification, obtained by the use of a semiconductor optical amplifier (SOA), which is polarization sensitive. Only the availability of a passive birefringence compensation circuit [6] allowed to overcome the limitations and to reach significant transmission performance [7].

A further example comes from the field of optical sensors, namely, the fiber-optic interferometric sensors. It is well known that fiber-optic interferometers (either with Mach–Zehnder or Michelson architecture) require stabilization of the state of polarization of the light arriving from the two arms. When one arm is the optical fiber itself, the presence of birefringence fluctuations perturbs the state of polarization leading to a severe impairment in the detected interference signal. The use of a suitable passive birefringence compensation circuit [8] had permitted to solve the problem and to make this kind of sensor available for in-field applications.

A last example of the effectiveness of passive birefringence compensation circuits is given by quantum cryptography. This technique is largely based on the Bennett and Brassard scheme that involves the generation of “qubits” based on the state of polarization [9]. Hence, the transmission through a birefringent medium can completely destroy the code. In 1997 a retracing circuit offering passive birefringence compensation was proposed [10] and remains the main solution for this technology.

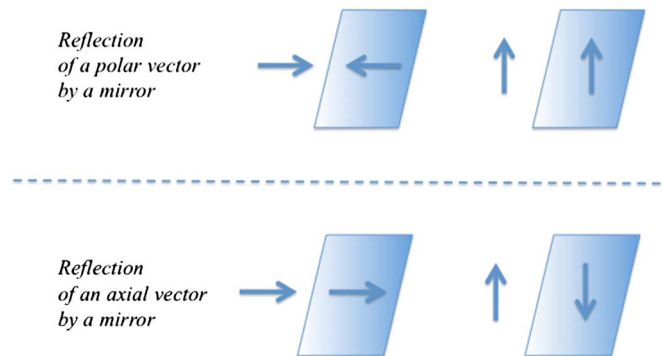
In summary, the compensation of unwanted birefringence presents many applications, from optical communications to optical sensing and quantum optics. To date, the most widely adopted technique is based on retracing schemes deploying a mirrored Faraday rotator (MFR), which compensates for linear and circular reciprocal birefringence. In general, optical circuits are also affected by circular nonreciprocal birefringence, which very recently [11] was proved to be compensated for by the mirrored quarter-wave plate (MQW) inserted in a retracing scheme. The same retracing scheme used with a common mirror permit compensating for the linear nonreciprocal birefringence.

The present article is aimed at giving a thorough analysis of the birefringence compensation issue in retracing circuits, using the mathematical tools of Stokes parameters, the Poincaré sphere, and Jones formalism. The physical mechanisms leading to the different kinds of birefringence, which can be classified as linear or circular, and also as reciprocal or nonreciprocal, will be reported. A complete review of the different mirroring devices, which are able to compensate for each specific birefringence, will be carried out, in view of giving a unified interpretation of the birefringence compensation.

2. SPIN, STOKES PARAMETERS, AND POINCARÉ SPHERE

An important feature of the photon spin is its representation as an *axial* vector. Since the photon spin is an intrinsic angular momentum it must be actually represented by an axial vector and not by a *polar* vector (the difference by the two types of vectors is masterfully explained in the books of Altmann [12] and Nye [13]). Although the representation of the two vectors is often the same, their properties are very different, particularly in connection with *retracing circuits*, where there is the presence of a mirror. For a polar vector, the component parallel to the mirror and its reflection are the same, while the component perpendicular and its reflection are opposite. In contrast, for an axial vector (also called a pseudo-vector), the component parallel to the mirror is changed by sign, while the component perpendicular to the mirror is left unchanged. Hence, when we represent a photon with spin $+1$ that hits a mirror, the returning beam is characterized by the representation in Fig. 1, and we say that “the photon reverses the spin, which becomes -1 (with respect to the propagation wave vector).” The polar vectors are related to the translations in space, and examples are the linear momentum, the propagation wave vector, and the electric field. It is the

Figure 1



Although the polar vector and the axial vector are presented by the same icon, they behave very differently when reflected by a mirror.

most usual kind of vector. The axial vectors are related to the rotations, having as examples the angular momentum and the magnetic field. A not obvious behavior characterizes the vector product between two polar vectors, which produces an axial vector, while the vector product between an axial and a polar vector produces a polar vector. The different behavior of polar and axial vectors in the presence of a mirror will be used at the end of Section 4 for explaining the difference between natural optical activity and the Faraday effect, the former being an effect of reciprocal circular birefringence, while the latter one of a nonreciprocal circular birefringence.

Although the spin is a physical entity and produces mechanical effects (as shown for the first time by the famous experiment carried out by Beth in 1936 [14]), we should shy away from attributing to the spin a naïve picture of a mechanical kind. To avoid mistakes, it is better to consider it an abstract entity: it is simply an intrinsic angular momentum. Because the spin is a measurable quantity, it is an observable. Fortunately, there is a connection between the classical treatment of polarized light and the quantum treatment of the spin, and this connection is represented by the Stokes parameters and the Poincaré sphere. As observed by Fano in [15]: “*The quantum theoretical treatment of the electromagnetic radiation fits in very well with the treatment of optics by means of the Stokes parameters. This is quite natural since the Stokes parameters are actually the observable quantities in phenomenological optics.*”

Following McMaster [16,17], we may consider a generic state of polarization of a light beam $|\psi\rangle$ as the linear combination of only two states, $|\psi_1\rangle$ and $|\psi_2\rangle$, since the spin presents only two eigenvalues:

$$|\psi\rangle = a_1|\psi_1\rangle + a_2|\psi_2\rangle, \quad (4)$$

where we have adopted Dirac's notation, which represents a quantum state as a ket $|\rangle$. In analogy with the spin of the particles, it is possible to fully describe the state of polarization as a bidimensional *spinor*. The mathematical concept of spinor was introduced by Cartan in 1913 [18]. For particles with two states of spin, as electrons and even photons [19,20], *three* different observables (represented by the Pauli matrices) *plus* the observable of the intensity (represented by the identity matrix) are needed for complete characterization of the state of polarization. Hence, we can obtain the following expectation values of the above four observables:

$$\begin{aligned} S_0 &= \langle\psi|\sigma_0|\psi\rangle = \begin{bmatrix} a_1^* & a_2^* \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a_1a_1^* + a_2a_2^*, \\ S_1 &= \langle\psi|\sigma_1|\psi\rangle = \begin{bmatrix} a_1^* & a_2^* \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a_1a_1^* - a_2a_2^*, \\ S_2 &= \langle\psi|\sigma_2|\psi\rangle = \begin{bmatrix} a_1^* & a_2^* \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a_1a_2^* + a_2a_1^*, \\ S_3 &= \langle\psi|\sigma_3|\psi\rangle = \begin{bmatrix} a_1^* & a_2^* \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = i(a_1a_2^* - a_2a_1^*), \end{aligned} \quad (5)$$

where σ_1 , σ_2 , and σ_3 are the Pauli matrices and S_0 , S_1 , S_2 , and S_3 are historically known as *Stokes parameters* of the polarization and identify a Stokes vector (given also in other notation, see, as an example, Clark and Grainger [21]):

$$S = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} = \begin{bmatrix} I \\ M \\ C \\ S \end{bmatrix} = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix}. \quad (6)$$

The Stokes parameters are the possible results of the measurement of the polarization of the beam obtained by orienting in a different mode a linear or a circular analyzer (see [21] and Fig. 2).

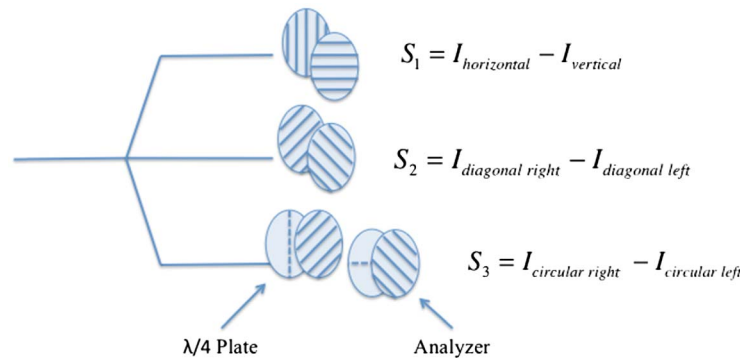
A very powerful geometric representation of the polarization is obtained by considering a Cartesian reference frame with axes S_1 , S_2 , and S_3 , in order that any point of coordinates corresponding to the Stokes parameters (S_1, S_2, S_3) represents a state of polarization. The Stokes parameter S_0 , which is the observable corresponding to the optical intensity, is related to the other three Stokes parameters, in case of a fully polarized state (i.e., with degree of polarization equal to 1), by the following relation [21]:

$$S_1^2 + S_2^2 + S_3^2 = S_0^2. \quad (7)$$

Assuming the usual normalization condition $S_0 = |a_1|^2 + |a_2|^2 = 1$, the state of polarization corresponding to the three Stokes parameters (S_1, S_2, S_3) can be naturally described by a point of a sphere of unit radius, known as a *Poincaré sphere*, as shown in Fig. 3. In this normalized case, the values of S_1 , S_2 , and S_3 range from -1 to $+1$, the extremes being in correspondence to the eigenstates of the respective Pauli matrices. Conventionally, we call these states

$$\begin{aligned} S_1 = \langle \sigma_1 \rangle = +1 &\leftrightarrow \text{horizontal linear state } |H\rangle, \\ S_1 = \langle \sigma_1 \rangle = -1 &\leftrightarrow \text{vertical linear state } |V\rangle, \\ S_2 = \langle \sigma_2 \rangle = +1 &\leftrightarrow \text{diagonal linear state } |Q\rangle, \\ S_2 = \langle \sigma_2 \rangle = -1 &\leftrightarrow \text{anti-diagonal linear state } |-Q\rangle, \\ S_3 = \langle \sigma_3 \rangle = +1 &\leftrightarrow \text{right circular state } |R\rangle, \\ S_3 = \langle \sigma_3 \rangle = -1 &\leftrightarrow \text{left circular state } |L\rangle. \end{aligned} \quad (8)$$

Figure 2



By means of an analyzer and a quarter-wave plate it is possible to recover all three Stokes parameters.

We remark that the above denominations (linear, horizontal, and so on) are *historical notations* and do not reflect (improper) geometrical representation of the photon spin but, instead, refer to the orientation of the optical field representation of the light beam under observation. In more detail, the linear states of polarization are characterized by an electric field oscillating as a function of time along a fixed direction, orthogonal to the propagation direction. The orientation of this oscillation corresponds to the polarization azimuth angle (which, for instance, is 0° for the horizontal state, 90° for the vertical state, and $\pm 45^\circ$ for the diagonal/anti-diagonal states). On the other hand, the right/left circular states are characterized by an electrical field with an arrow describing a circle, as a function of time, in a plane orthogonal to the propagation direction, with right-/left-handedness.

The above six main states of Eq. (5) are the intersection points of the Poincaré sphere with the S_1 , S_2 , and S_3 axes. Moreover, each point on the Poincaré sphere can also be identified by using spherical angular coordinates 2θ (angle on the equatorial plane or longitude) and 2χ (elevation angle or latitude). So we have [21]

$$S_1 = S_0 \cos(2\chi) \cos(2\theta), \quad S_2 = S_0 \cos(2\chi) \sin(2\theta), \quad S_3 = S_0 \sin(2\chi), \quad (9)$$

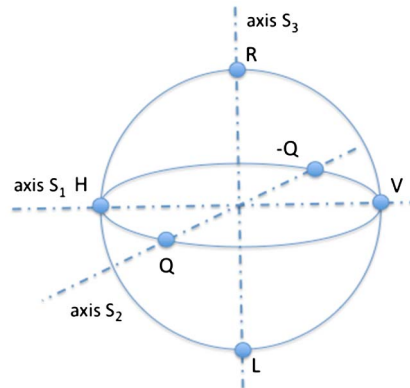
and also

$$\tan(2\theta) = \frac{S_2}{S_1}, \quad \sin(2\chi) = \frac{S_3}{S_0}. \quad (10)$$

Furthermore, the angle θ represents the polarization azimuth, that is, the orientation of the major axis for a general elliptical state, while χ is the ellipticity. The linear states of polarization have ellipticity equal to zero and correspondingly lay on the equator of the Poincaré sphere. The north hemisphere corresponds to the right elliptical states, with positive ellipticity, while the south hemisphere corresponds to the left elliptical states. In particular, the right/left circular states, corresponding to the north/south poles, have ellipticity equal to $\pm 45^\circ$, respectively.

Finally, the correspondence between the Stokes parameters, the electromagnetic picture, the Pauli operators, and the Poincaré sphere representation is sketched in Fig. 4.

Figure 3



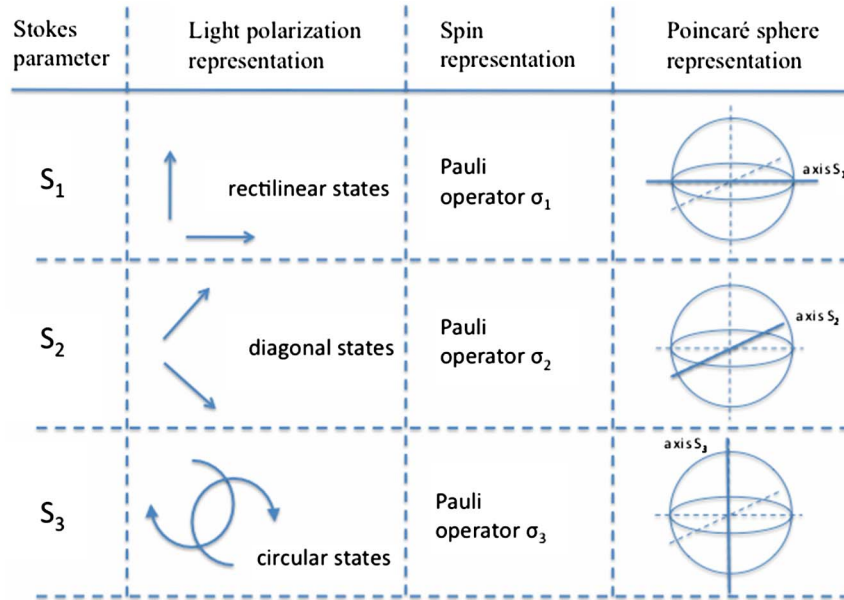
Poincaré sphere.

3. JONES REPRESENTATION

Although the Stokes parameters allow complete description of any state and degree of polarization, including the partially polarized beams, characterized by $S_1^2 + S_2^2 + S_3^2 < S_0^2$ (see as examples [22,23]) and even the polarization evolution through whatever optical component (including the birefringence devices), it is customary to follow the evolution of the state of polarization by using a more friendly representation using the Jones vectors (so named because they were proposed by Jones in 1941 [24]). The Jones vectors can be connected to the photon spin by using again Eqs. (5) and (7). In fact, when we have introduced in Eq. (7) the six states corresponding to the eigenvalues of the Pauli operators, we left unexpressed the six respective eigenstates. It is easy to demonstrate that Eq. (7) can be completed in the following way:

$$\begin{aligned}
 \langle H | \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} | H \rangle &= \langle \sigma_1 \rangle = +1 \quad \text{with } |H\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\
 \langle V | \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} | V \rangle &= \langle \sigma_1 \rangle = -1 \quad \text{with } |V\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\
 \langle Q | \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} | Q \rangle &= \langle \sigma_2 \rangle = +1 \quad \text{with } |Q\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \\
 \langle -Q | \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} | -Q \rangle &= \langle \sigma_2 \rangle = -1 \quad \text{with } |-Q\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \\
 \langle R | \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} | R \rangle &= \langle \sigma_3 \rangle = +1 \quad \text{with } |R\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}, \\
 \langle L | \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} | L \rangle &= \langle \sigma_3 \rangle = -1 \quad \text{with } |L\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix}.
 \end{aligned} \tag{11}$$

Figure 4



Equivalent representation of the Stokes parameters, light polarization, photon spin, and Poincaré sphere.

The two eigenstates of the Pauli operator σ_1 , that is, the horizontal and vertical states, are the natural basis states of the Jones representation, which expresses the states of polarization as a bidimensional vector J with components given by the horizontal and vertical components of the electrical field:

$$J = \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} E_{x0} \exp(i\delta_{x0}) \\ E_{y0} \exp(i\delta_{y0}) \end{bmatrix}, \quad (12)$$

where

$$\begin{aligned} E_{x0} &= \text{amplitude of the field component along the axis } x, \\ E_{y0} &= \text{amplitude of the field component along the axis } y, \\ \delta_{x0} &= \text{initial phase of the field component along the axis } x, \\ \delta_{y0} &= \text{initial phase of the field component along the axis } y, \end{aligned} \quad (13)$$

and the time dependence of the phase has been omitted. Moreover, we can say that the states of polarization are identified by the Jones vectors, apart from a complex multiplicative coefficient. Actually, the product of the Jones vector by a complex number gives the same variation of amplitude and phase for both field components, but maintains unchanged the state of polarization.

We can obtain the field representation, in terms of Jones vectors, for the six main states of Eq. (11), taking into account the specific relations between the field components. The horizontal and vertical states are characterized by $E_{y0} = 0$ and $E_{x0} = 0$, respectively. Hence we can write

$$\begin{aligned} |H\rangle &\leftrightarrow \begin{bmatrix} E_{x0} \exp(i\delta_{x0}) \\ 0 \end{bmatrix} = E_{x0} \exp(i\delta_{x0}) \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ |V\rangle &\leftrightarrow \begin{bmatrix} 0 \\ E_{y0} \exp(i\delta_{y0}) \end{bmatrix} = E_{y0} \exp(i\delta_{y0}) \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned} \quad (14)$$

The diagonal and anti-diagonal states are characterized by $E_{y0} = E_{x0} = E_0/\sqrt{2}$ and either $\delta_{y0} = \delta_{x0}$ or $\delta_{y0} = \delta_{x0} + \pi$; hence

$$\begin{aligned} |Q\rangle &\leftrightarrow \begin{bmatrix} E_{x0} \exp(i\delta_{x0}) \\ E_{x0} \exp(i\delta_{x0}) \end{bmatrix} = \frac{E_0 \exp(i\delta_{x0})}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \\ |-Q\rangle &\leftrightarrow \begin{bmatrix} E_{x0} \exp(i\delta_{x0}) \\ E_{x0} \exp[i(\delta_{x0} + \pi)] \end{bmatrix} = \frac{E_0 \exp(i\delta_{x0})}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \end{aligned} \quad (15)$$

The right and left circular states are characterized by $E_{y0} = E_{x0} = E_0/\sqrt{2}$ and either $\delta_{y0} = \delta_{x0} + \pi/2$ or $\delta_{y0} = \delta_{x0} - \pi/2$; hence

$$\begin{aligned} |R\rangle &\leftrightarrow \begin{bmatrix} E_{x0} \exp(i\delta_{x0}) \\ E_{x0} \exp[i(\delta_{x0} + \frac{\pi}{2})] \end{bmatrix} = \frac{E_0 \exp(i\delta_{x0})}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}, \\ |L\rangle &\leftrightarrow \begin{bmatrix} E_{x0} \exp(i\delta_{x0}) \\ E_{x0} \exp[i(\delta_{x0} - \frac{\pi}{2})] \end{bmatrix} = \frac{E_0 \exp(i\delta_{x0})}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix}. \end{aligned} \quad (16)$$

We point out that the above Jones vectors of the six main polarization states are equal, apart from a complex multiplicative coefficient, to the vectors [Eq. (11)] of

the quantum representation. Of course, other vectors can be obtained by choosing either a “diagonal” or “circular” basis (that is, the eigenstates of either σ_1 or σ_2 as basis states), but the Jones vectors in the “rectilinear” basis, shown by Eqs. (14)–(16), are the most popular. Moreover, the intensity of the field represented by the Jones vector is by definition

$$I = \begin{bmatrix} E_{x0} \exp(i\delta_x) & E_{y0} \exp(i\delta_y) \end{bmatrix}^* \begin{bmatrix} E_{x0} \exp(i\delta_x) \\ E_{y0} \exp(i\delta_y) \end{bmatrix} = (E_{x0})^2 + (E_{y0})^2, \quad (17)$$

which is by definition the Stokes parameter S_0 . More in general, from the Stokes parameter definition given by Eq. (5) and the Jones vector expression of Eq. (12), we can write the Stokes parameters as a function of the field components as [22]

$$\begin{aligned} S_0 &= [(E_{x0})^2 + (E_{y0})^2], \\ S_1 &= [(E_{x0})^2 - (E_{y0})^2], \\ S_2 &= [2E_{x0}E_{y0} \cos(\delta_y - \delta_x)], \\ S_3 &= [2E_{x0}E_{y0} \sin(\delta_y - \delta_x)]. \end{aligned} \quad (18)$$

The above equations permit connecting the Jones representation (which is essentially a field representation) with the Stokes representation (which is essentially an intensity representation) of the polarization.

If a state of polarization is represented by the Jones vector [Eq. (12)], then any optical device transforming the polarization can be considered an operator represented by a matrix (said *Jones matrix*). A rich literature permitted determination of the Jones matrix for whatever significant device operating on the polarization (such as the analyzer of polarization, mirror, beam splitter, or optical plate with birefringence) and calculation of the polarization evolution for the most complex optical circuits involving polarization. The algebra of these circuits follows the rules of the matrix calculus.

- Any element represented by a Jones matrix M transforms the input Jones vector J_{in} into an output Jones vector J_{out} according to the usual product row by column:

$$J_{\text{out}} = MJ_{\text{in}}. \quad (19)$$

- If we have a cascade of elements with Jones matrices $M_1, M_2, \dots, M_n$, the overall effect can be summarized by the product of the respective Jones matrices ordered in the inverse sequence from left to right:

$$J_{\text{out}} = M_n \dots M_2 M_1 J_{\text{in}}. \quad (20)$$

4. BIREFRINGENCE

When a material loses isotropy, that is, it becomes anisotropic, the simple scalar relation between the *optical field vector* $\mathbf{E}$ of the beam propagating into the material and the *displacement vector* $\mathbf{D}$ must be expressed by a *permittivity tensor*. It assumes two different expressions for the nonactive and active materials [22]. We point out that the effects giving a nonlinear dependence of the birefringence, i.e., the Kerr effect and the Cotton–Mouton effect, are neglected because they involve more than one wave and require a specific treatment for the state of polarization of the different components.

4.1. Nonactive Materials

In the case of nonactive materials, the tensorial relation between vectors $\mathbf{D}$ and $\mathbf{E}$ is expressed as

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \varepsilon_0 \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}, \quad (21)$$

where ε_0 is the vacuum permittivity and ε_{ij} are the relative permittivity coefficients. All the coefficients ε_{ij} are real and the relation $\varepsilon_{ij} = \varepsilon_{ji}$ holds. Hence, the permittivity tensor in Eq. (21) becomes symmetric and the independent dielectric coefficients are only six. Generalizing the scalar relation $n^2 = \varepsilon$ to the case of anisotropic material, we can introduce the squared refractive index coefficients $n_{ij}^2 = \varepsilon_{ij}$, and, by substituting in Eq. (21), we can write

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \varepsilon_0 \begin{bmatrix} n_{11}^2 & n_{12}^2 & n_{13}^2 \\ n_{12}^2 & n_{22}^2 & n_{23}^2 \\ n_{13}^2 & n_{23}^2 & n_{33}^2 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}. \quad (22)$$

The squared refractive index coefficients n_{ij}^2 define the *index ellipsoid*, which provides a complete representation of the birefringence properties for any direction of propagation of the optical beam. In general, by means of a rotation of the reference frame, it is always possible to diagonalize the tensor in Eq. (22) and obtain a relation of the type

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \varepsilon_0 \begin{bmatrix} n_x^2 & 0 & 0 \\ 0 & n_y^2 & 0 \\ 0 & 0 & n_z^2 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}. \quad (23)$$

The three axes x , y , and z of this diagonalized form are the principal axes, with corresponding principal indices n_x , n_y , and n_z . Depending on the specific crystal symmetry of the material, we have the following three cases (see as examples [22,23,25]).

- For isotropic crystals (the crystals of the cubic class) and amorphous materials (the glasses), the index ellipsoid is a sphere and all the principal indices have the same value. So Eq. (23) has the simple expression

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \varepsilon_0 \begin{bmatrix} n_0^2 & 0 & 0 \\ 0 & n_0^2 & 0 \\ 0 & 0 & n_0^2 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}, \quad (24)$$

where all the principal indices have the same value n_0 .

- For uniaxial crystals (crystals of the tetragonal, hexagonal, and trigonal classes), the index ellipsoid is an ellipsoid of revolution around a principal axis and Eq. (23) becomes

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \varepsilon_0 \begin{bmatrix} n_o^2 & 0 & 0 \\ 0 & n_e^2 & 0 \\ 0 & 0 & n_o^2 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}, \quad (25)$$

where n_o is the ordinary index and n_e the extraordinary index.

- For biaxial crystals (the crystals of triclinic, monoclinic, and orthorhombic classes), the three principal indices have different values, as in the more general case described by Eq. (23).

The major part of the materials, which are used for the components in the optical polarization circuits, are either isotropic or uniaxial. The isotropic materials, characterized by a single value n_0 of the refractive index, as shown by Eq. (24), do not possess any birefringence. Uniaxial crystals (like quartz and calcite) show two different behaviors. In case of propagation of the light along the *optical axis* given by the revolution axis of the index ellipsoid, assumed as axis y according to Eq. (25), the refraction index has the same value n_0 for any transverse direction of the electric field, and the crystal behaves like an isotropic crystal with no birefringence. Otherwise, for a propagation direction different from the optical axis, there are two different values of refractive index (i.e., the phenomenon of *double refraction* or *birefringence*) for two orthogonal linear states of polarization. The maximum effect of birefringence in a uniaxial crystal is for a direction of propagation aligned with a principal axis orthogonal to the optical axis. For instance, in the case of propagation along axis z , considering the diagonalized tensor in Eq. (24), there are the two different refraction indices, n_o and n_e , in correspondence with the two linearly polarized states with electric fields aligned to axes x and y , respectively. Hence, we have a birefringence of an amount given by the following difference between the refractive indices:

$$\Delta n = n_y - n_x = n_e - n_o, \quad (26)$$

which can be negative (e.g., for the calcite) or positive (e.g., for the quartz).

The situation is more complicated for the biaxial crystals. According to Eq. (23), there are three different principal indices, so in the case of propagation along a principal axis, there is a birefringence due to the difference between the principal indices corresponding to the other two principal axes. In general, an optical axis of a crystal is defined as a direction of propagation through the crystal that does not have birefringence. Therefore, light propagates along an optical axis like in an isotropic medium. Biaxial crystals are so called because they have two optical axes, different from the principal axes, while uniaxial crystals have a single optical axis, given by the principal axis representing the revolution axis of the index ellipsoid.

The above-described anisotropic nonactive materials (either uniaxial or biaxial) exhibit birefringence in the case of propagation not directed along an optical axis. In this case, there are two directions, both orthogonal to the propagation direction, that identify the birefringence axes, characterized by the vectors $\mathbf{E}$ and $\mathbf{D}$ remaining parallel during propagation. The two orthogonal linear states of polarization with an electric field aligned to the birefringence axes are not changed, but acquire a mutual phase difference, due to the birefringence, equal to

$$\phi = k_0 \Delta n L = \frac{2\pi}{\lambda_0} \Delta n L, \quad (27)$$

where $\lambda_0 = 2\pi/k_0$ is the wavelength in vacuum and L is the length of propagation through the birefringent material. We now consider the propagation along axis z and the birefringence axes aligned to axes x and y , with respective indices n_x and n_y . If $n_x < n_y$, then axis x is the fast axis (while axis y is the slow axis), and, according to Eqs. (26) and (27), we have positive values for birefringence Δn and phase difference ϕ . The Jones matrix of such a birefringent element, also called a *retarder plate*, with the above choice of Cartesian reference frame so that the birefringence axes are horizontal (fast axis) and vertical (slow axis), can be written as

$$R[0, \phi] = \begin{bmatrix} \exp(+i\phi/2) & 0 \\ 0 & \exp(-i\phi/2) \end{bmatrix}, \quad (28)$$

where it is supposed that $\phi/2$ is the phase advance accumulated by the horizontal component E_x , and $-\phi/2$ is the phase delay accumulated by the vertical component E_y . The optical component acting on the polarization according to the above Jones matrix $R[0, \phi]$ is called an *optical retarder*, oriented at an angle 0 (being horizontal the fast axis) and with phase retardation ϕ .

It can be easily verified that the eigenvectors of the Jones matrix, in the diagonal form expressed by Eq. (28), are the Jones vectors corresponding to the horizontal and vertical states of polarization, with respective eigenvalues $\exp(\pm i\phi/2)$. Thus the horizontal and vertical states are the orthogonal eigenstates of the retarder, that is, the only states of polarization remaining unchanged. In fact, the Jones vectors of the eigenstates of the retarder are just multiplied by the phase factors given by the respective eigenvalues. Moreover, we will see in Section 5 how to obtain the more general Jones matrix $R[\gamma, \phi]$ for a retarder with fast axis oriented at an angle γ . In this case, the eigenstates are the orthogonal *linear* states of polarization with electric fields respectively aligned to the fast and slow axes of birefringence. Therefore, this kind of birefringence, introduced by nonactive materials, is also called *linear*.

The linear birefringence can be *natural*, that is, possessed by the material itself due to the specific crystal class (like the quartz), or even *induced* by external agents that perturb the isotropy of the material. We may have three kinds of agents causing a linear birefringence of amount $\Delta n(n_y - n_x)$ directly proportional to the amplitude of the agent itself: mechanic, electric, and magnetic. Hence we have:

- (a) opto-mechanical linear birefringence (also known as the elasto-optic effect);
- (b) electro-optical linear birefringence (also known as the Pockels' effect); and
- (c) magneto-optical linear birefringence (also known as nonreciprocal linear birefringence; see, for example, [26–28]).

4.2. Active Materials

In the case of active materials, the tensorial relation between the vectors $\mathbf{D}$ and $\mathbf{E}$ assumes, in a suitable Cartesian reference, the following form [22]:

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \epsilon_0 \begin{bmatrix} \epsilon_x E_x + i(\boldsymbol{\alpha} \times \mathbf{E})_x \\ \epsilon_y E_y + i(\boldsymbol{\alpha} \times \mathbf{E})_y \\ \epsilon_z E_z + i(\boldsymbol{\alpha} \times \mathbf{E})_z \end{bmatrix}, \quad (29)$$

where we point out the presence of a *vector product* (with symbol $\times$) between the vector $\boldsymbol{\alpha}$, which depends on the material, and the vector $\mathbf{E}$, which represents the optical field. By developing the second member of Eq. (29), we can write

$$\begin{aligned} \begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} &= \epsilon_0 \begin{bmatrix} \epsilon_x & -i\alpha_z & i\alpha_y \\ i\alpha_z & \epsilon_y & i\alpha_x \\ -i\alpha_y & -i\alpha_x & \epsilon_z \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \\ &= \left\{ \epsilon_0 \begin{bmatrix} n_x^2 & 0 & 0 \\ 0 & n_y^2 & 0 \\ 0 & 0 & n_z^2 \end{bmatrix} + i\epsilon_0 \begin{bmatrix} 0 & -\alpha_z & \alpha_y \\ \alpha_z & 0 & \alpha_x \\ -\alpha_y & -\alpha_x & 0 \end{bmatrix} \right\} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}. \end{aligned} \quad (30)$$

We see that the permittivity tensor is constituted by two addenda. The first one is a diagonal (hence, symmetric) tensor responsible for a linear birefringence, equivalent to the case of non-active materials as shown by Eq. (23), while the second addendum is an anti-symmetric tensor characteristic of optically active materials. The description

of the polarization evolution through active materials is more complicated than in the case of nonactive materials and depends on the particular forms of the permittivity tensor for the different crystalline classes [25]. For instance, all the centro-symmetric crystals and the amorphous materials, in the absence of external agents, do not present optical activity.

Now we consider a typical condition for active materials of having the gyration vector α aligned with the propagation direction along axis z . Substituting $\alpha_x = \alpha_y = 0$ in Eq. (30), together with the condition $n_x = n_y = n_0$, meaning that the symmetric part of the permittivity tensor does not introduce any birefringence, we obtain

$$\begin{aligned} \begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} &= \varepsilon_0 \begin{bmatrix} n_0^2 & -i\alpha_z & 0 \\ i\alpha_z & n_0^2 & 0 \\ 0 & 0 & n_z^2 \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \\ &= \left\{ \varepsilon_0 \begin{bmatrix} n_0^2 & 0 & 0 \\ 0 & n_0^2 & 0 \\ 0 & 0 & n_z^2 \end{bmatrix} + i\varepsilon_0 \begin{bmatrix} 0 & -\alpha_z & 0 \\ \alpha_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}. \end{aligned} \quad (31)$$

It can be demonstrated that an optically active material described by Eq. (31) induces a rotation around axis z of a linearly polarized electric field oscillating along a direction orthogonal to axis z , that is, a *rotation of polarization*. The action of this polarization rotator is expressed by the Jones matrix:

$$C[\delta] = \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix}, \quad (32)$$

where the angle of rotation δ depends on the gyration vector component α_z . We point out that a polarization rotator varies the azimuth θ of any input polarization state by an amount equal to the above-introduced rotation angle δ , while leaving unchanged the ellipticity χ . It is worth remarking that the eigenvectors of the rotation matrix [Eq. (32)] are the Jones vectors of the right and left circular states, with respective eigenvalues $\exp(\mp i\delta)$. Therefore, the rotation of polarization by an angle δ induced by the optical activity is equivalent to produce a phase difference 2δ between the two circular eigenstates. We can also define two different indices of refraction, n_+ and n_- , corresponding to the right and left circular states, respectively, and given by

$$n_{\pm} = (n_0^2 \pm \alpha_z)^{1/2}, \quad (33)$$

as can be derived from Eq. (31). Assuming the typical condition $\alpha_z \ll n_0^2$, we can use the approximated values

$$n_{\pm} \cong n_0 \pm \frac{\alpha_z}{2n_0}. \quad (34)$$

We can say that the optical activity gives a *circular birefringence*, that is, a difference between the circular refraction indices, of amount

$$\Delta n_{\text{circ}} = n_+ - n_- \cong \frac{\alpha_z}{n_0}. \quad (35)$$

Considering the same kind of relation between phase difference and index difference for both linear and circular birefringence, we can substitute $\phi = 2\delta$ and $\Delta n = \Delta n_{\text{circ}}$ in Eq. (27), obtaining

$$2\delta = k_0 \Delta n_{\text{circ}} L = \frac{2\pi}{\lambda_0} \Delta n_{\text{circ}} L \quad (36)$$

for a propagation of length L through an optically active material. Owing to Eq. (35), the rotation angle δ has the following dependence on the gyration vector component α_z along the propagation direction:

$$\delta = \frac{k_0}{2n_0} \alpha_z L = \frac{\pi}{\lambda_0 n_0} \alpha_z L. \quad (37)$$

This rotation angle δ is positive for $\Delta n_{\text{circ}} = n_+ - n_- > 0$, corresponding to $n_- < n_+$, that is, the left circular state is the fast eigenstate, while the right circular state is the slow eigenstate.

The value of this component α_z is determined by the material structure and by the external agents. In fact, the circular birefringence can be *natural*, thus possessed by the material itself due to the specific structure (e.g., the crystal of quartz or solutions of chiral molecules), or even *induced* by external agents that perturb the intrinsic symmetry of the material. We may have three kinds of agents causing a circular birefringence of amount $\Delta n_{\text{circ}} = (n_+ - n_-)$ directly proportional to the amplitude of the agent itself. Hence we have:

- (d) opto-mechanical circular birefringence (also known as the elasto-optic effect);
- (e) electro-optical circular birefringence (also known as the electrogyration effect [29]); and
- (f) magneto-optical circular birefringence (also known as the Faraday effect).

We now introduce the theme of reciprocity, which will be discussed more in detail in the next section. For a typical circular birefringence induced by natural optical activity, the gyration vector α is aligned to the wave vector $\mathbf{k}$ and both vectors are polar. This means that an inversion of the reference axis z causes a change of sign of the component α_z in the new reference frame. Equivalently, according to Fig. 1, an inversion of the propagation direction (along axis z), due to a physical effect of mirroring, corresponds to an inversion of the polar vector α and a change of sign of α_z (relative to the initial reference frame before mirroring). The polarization rotation angle δ , directly proportional to α_z for Eq. (37), changes sign after mirroring (maintaining fixed the reference frame), as well. Hence, the polarization rotation is cancelled by the re-tracing. We also remark that the vector product $\alpha \times \mathbf{E}$, responsible for the optical activity, is in this case an axial vector orthogonal to the propagation direction and, hence, is reversed by mirroring (see Fig. 1), confirming the cancellation of the effect. In conclusion, we can state that the circular birefringence due to the natural optical activity, as an example of *reciprocal* circular birefringence, is compensated for by a common mirror.

A different situation occurs for the *nonreciprocal* circular birefringence due to the optical activity induced by the magneto-optical Faraday effect. In this case, the gyration vector α is an axial vector directly proportional to the magnetic flux density $\mathbf{B}$ and is not reversed by mirroring, according to Fig. 1. Hence, the signs of both component α_z and rotation angle δ are not changed by mirroring (keeping the reference frame fixed), causing a doubling of the polarization rotation effect. Moreover, we can note that the vector product $\alpha \times \mathbf{E}$ is in this case a polar vector orthogonal to the propagation direction and, hence, is not reversed by mirroring (see Fig. 1), confirming the doubling of the effect. Finally, we can say that the nonreciprocal circular birefringence, as in the above example of optical activity induced by the Faraday effect, is not

compensated for by a common mirror. We will see in Section 6 how to compensate for such birefringence.

5. EVOLUTION OF THE STATE OF POLARIZATION IN A RETRACING CIRCUIT

In the previous section we have seen that the effects of birefringence can be described by means of two kinds of optical components acting on the polarization: the first one is the *retarder*, corresponding to linear birefringence, whose Jones matrix $R[0, \phi]$ in the case of birefringence axes aligned with the transverse Cartesian axes x and y is given by Eq. (28); the second one is the *rotator*, corresponding to circular birefringence, whose Jones matrix $C[\delta]$ is given by Eq. (32). We now have the fundamental elements that permit us to face the problem of shaping the evolution of the state of polarization of a light beam through a retracing circuit, containing optical components with a generic birefringence and a mirror that intercepts the beam and inverts the propagation direction (as schematized by Fig. 5). In the following we adopt the Jones representation for retracing beams introduced by Bhandari [30] and the Poincaré sphere representation reported by Wright [31].

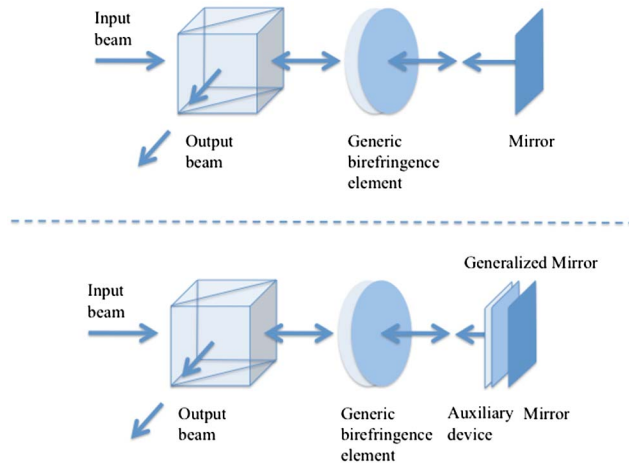
First, let us analyze what happens when a retarder is not considered in a reference frame with axes x and y aligned with the birefringence axes, but is considered in a rotated frame by an angle γ around the propagation axis z . Since the rotation does not perturb the entity of the phase retardation ϕ , according to the well-known properties of the matrices in the case of rotation of the reference frame, we have that the Jones matrix of the retarder oriented at an angle γ with respect to the axis x is

$$R[\gamma, \phi] = C[\gamma]R[0, \phi]C[-\gamma], \quad (38)$$

where $C[\gamma]$ is the Jones matrix of a rotator by an angle γ . Evaluating the matrices $R[0, \phi]$ and $C[\gamma]$ by means of Eqs. (28) and (32), respectively, we obtain the general expression

$$R[\gamma, \phi] = \begin{bmatrix} \cos(\phi/2) + i \sin(\phi/2) \cos(2\gamma) & i \sin(\phi/2) \sin(2\gamma) \\ i \sin(\phi/2) \sin(2\gamma) & \cos(\phi/2) - i \sin(\phi/2) \cos(2\gamma) \end{bmatrix}. \quad (39)$$

Figure 5



Retracing optical circuit with a mirror (above) and a generalized mirror (below).

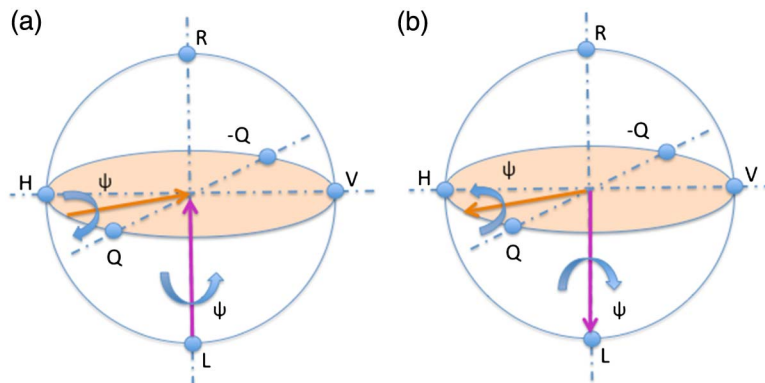
Such a mathematical expression of the action of a generic retarder is quite difficult to figure. Nevertheless, the Poincaré sphere representation allows us to overcome this problem and to offer a powerful visualization of the polarization evolution when crossing any kind of birefringence element by means of simple geometrical rules.

Then, let us consider the effect of a rotation of the reference frame on the Jones matrix $C[\gamma]$ of the rotator. The angle of rotation γ is the difference between the azimuth angles of the linear states of polarization at the exit and at the entrance of the rotator: clearly this difference is not varied by a rotation of the reference frame, and we can consider the Jones matrix of the rotator independent from the orientation of the reference frame, except for a not relevant multiplicative phase factor.

The Poincaré sphere representation is discussed in the following. The main states of polarization are represented on the sphere by the points shown in Fig. 6, corresponding to the points of intersection between the sphere and the respective axes S_1 , S_2 , and S_3 . In this frame, any generalized birefringence, giving a phase retardation ψ between two elliptical orthogonal eigenstates of polarization, impresses a rotation of the sphere by an angle ψ around an axis passing through the center of the sphere and the points corresponding to the eigenstates. The positive birefringence (i.e., positive ψ) is represented by a birefringence vector directed from the fast eigenstate to the center, with anti-clockwise rotation around this vector seen from the arrow [as shown in Fig. 6(a)]. If the birefringence is negative (i.e., negative ψ), the representing rotation vector is directed from the center to the fast eigenstate [as shown in Fig. 6(b)]. In particular, we will see in detail the action on the Poincaré sphere of both linear (i.e., with linear eigenstates) and circular (i.e., with circular eigenstates) birefringence.

Let us now introduce the rules for the *action of a linear birefringence plate* $R[\gamma, \varphi]$ with fast axis oriented at γ (an azimuthal angle that, without loss of generality, we consider positive when measured from point H toward point Q). Let us further suppose that the entrance state of polarization is represented by a generic point P on the Poincaré sphere. The action can be followed in a “polar representation” in Fig. 7, that

Figure 6



(a) Linear birefringence is represented by a vector lying on the equatorial plane of the Poincaré sphere that rotates clockwise around the sphere itself. Circular birefringence is represented by a vector whose direction coincides with the polar axis and rotates anti-clockwise around the sphere itself. (b) In the presence of negative birefringence, the vector is reversed.

is, the projection from the top of the sphere (the north pole corresponding to the right circular state R).

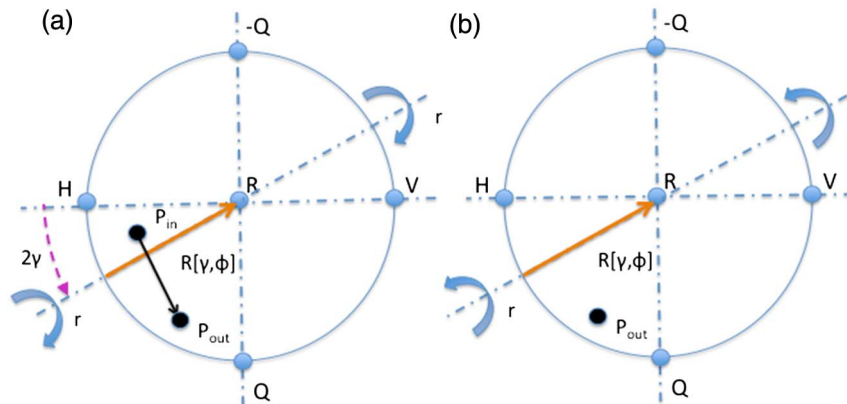
- Rule 1. Rotate the Poincaré sphere by an angle ϕ around the birefringence vector oriented at an angle 2γ . The point representing the entrance state of polarization follows rigidly the movement and traces an arc in a plane orthogonal to the direction of the rotation vector corresponding to the linear birefringence, moving from P_{in} to P_{out} , as shown in Fig. 7(a).
- Rule 2. Reset the movement of the sphere, but keep the point on the previous position P_{out} . This is the new state of polarization of the light after the birefringence element.

Similarly, let us now introduce the rules for the *action of a circular birefringence plate* described by the rotator Jones matrix $C[\delta]$, which introduces a rotation of the polarization azimuth by an angle δ , corresponding to a phase difference 2δ between circular eigenstates $C[\delta]$. A positive sign of δ means that the left circular state is the fast eigenstate, while the right circular state is the slow eigenstate. The action can be followed in “polar representation” on Fig. 8. A positive rotation corresponds to an anti-clockwise rotation around the polar axis, as seen from the north pole. Figure 8 shows such a movement along a parallel arc in the polar view of the Poincaré sphere. We remark that an azimuth rotation by an angle δ corresponds to a doubled rotation 2δ around the polar axis of the Poincaré sphere, according to the fact that a state with polarization azimuth θ is represented by a point on the Poincaré sphere with longitude 2θ .

- Rule 1. Rotate the Poincaré sphere by an angle 2δ around the polar axis. The point representing the entrance state of polarization follows rigidly the movement and traces a parallel arc, moving from P_{in} to P_{out} in Fig. 8(a).
- Rule 2. Reset the movement of the sphere, but keep the point on the previous position P_{out} . This is the new state of polarization of the light after the birefringence element.

To complete our representation for analyzing a retracing circuit, we need to represent the action of a mirror. According to Bhandari [30], for keeping reference axis z aligned

Figure 7



Action of the linear birefringence vector on a generic state of polarization P_{in} : (a) all the sphere, including P_{in} , is rotated around the axis $r-r$ by the angle γ , and P_{in} migrates in P_{out} ; (b) the sphere returns to the original position but leaving P_{out} in its position.

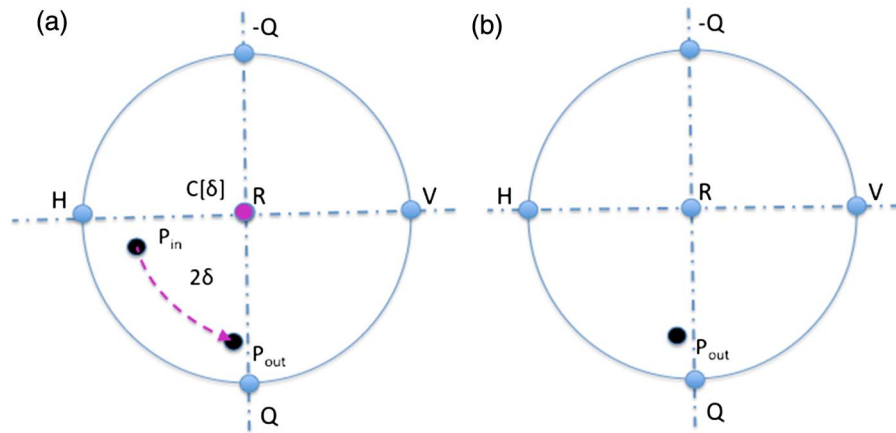
with the propagation direction, we consider an inversion of axis z on the returning beam after the mirror. Moreover, the original Cartesian right-handed reference frame remains right-handed after mirroring if another reference axis is inverted, like axis y without sake of generality, in addition to axis z . In the following, we assume that the reference frame after the mirror is obtained from the reference frame before the mirror, inverting axes y and z , while maintaining unchanged axis x .

We want now to describe the transformation of the states of polarization operated by the mirror, taking into account this reference change. We remark that the electric field of an optical beam is a polar vector orthogonal to the direction of propagation, thus it is not changed by the mirroring (according to Fig. 1). This means that the direction of oscillation of the electric field of a linearly polarized beam is not changed. However, the inversion of axis y changes the sign of the orientation angle of a transverse direction with respect to axis x (unchanged after mirroring). Hence the horizontal state is not changed by the mirror, as well as the vertical state (in fact, the inversion of axis y causes just a phase shift of π on a vertically oscillating field), while the *diagonal* state becomes *anti-diagonal* and *vice versa*. Regarding the action of the mirror on the circular states, we remark at first that the handedness of a circular state is related to the spin component along the propagation direction. Then we have that the mirror inverts the direction of propagation, but does not change the spin (being described by an axial vector orthogonal to the mirror; see again Fig. 1), hence it transforms the right circular state into the left one and *vice versa*. Finally, we can summarize the effect of the mirror on the main six states of polarization on the Poincaré sphere as

$$\begin{aligned} |H\rangle &\Rightarrow |H\rangle, & |V\rangle &\Rightarrow |V\rangle, \\ |Q\rangle &\Rightarrow |-Q\rangle, & |-Q\rangle &\Rightarrow |Q\rangle, \\ |R\rangle &\Rightarrow |L\rangle, & |L\rangle &\Rightarrow |R\rangle, \end{aligned} \quad (40)$$

which is equivalent to a rotation of π around the axis S_1 , shown in Fig. 9. Furthermore, this rotation is the same when describing a half-wave plate with horizontal and vertical axes [21]; hence, the Jones's matrix of the mirror is

Figure 8



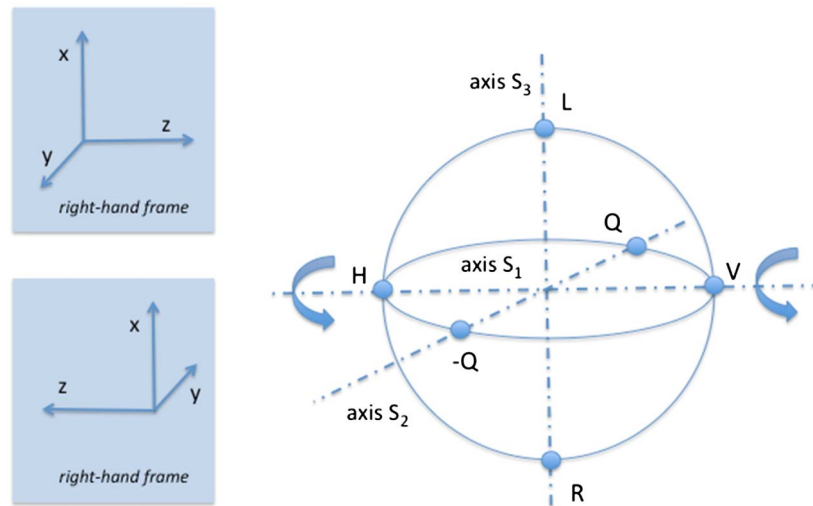
Action of the circular birefringence vector on a generic state of polarization P_{in} : (a) all the sphere, including P_{in} , is rotated around the polar axis by the angle 2δ , and P_{in} migrates in P_{out} ; (b) the sphere returns to the original position, but leaving P_{out} in its position.

$$M_{\text{mirror}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (41)$$

The inversion of propagation due to the presence of a mirror involves the principle of reciprocity. Following Brosseau [23]: “...we wish to address the issue of reciprocity constraint on polarization transformation matrices, specifically, the behaviour of the state of polarization in an optical system when the direction of propagation is reversed, that is when the incident and the scattered beams are interchanged. The principle of reciprocity (also termed the Helmholtz reciprocity theorem) in its general form states that the input and output of any linear passive optical system can be interchanged without altering the response of the optical system. An optical medium that satisfies the principle of reciprocity is termed a reciprocal medium; otherwise it is called a nonreciprocal medium. This reciprocity is a consequence of the time-reversal invariance requirement on Maxwell’s equation.” In general, the birefringent elements, illustrated in Section 4, can be either reciprocal or nonreciprocal, depending on the specific kind of interaction considered.

We start investigating the effect of the mirror on the linear birefringence, represented by the operator $R[\gamma, \phi]$, distinguishing between the reciprocal and nonreciprocal cases. We have seen that the inversion of the reference axis y after mirroring, for maintaining a right-hand Cartesian reference frame, changes the sign of the orientation angle of a transverse direction, so γ is transformed in $-\gamma$ by mirroring. Moreover, the characteristic effect of non-reciprocity on linear birefringence is to change the sign of the phase retardation ϕ in correspondence to the inversion of the propagation direction. Second, we analyze the effect of the mirror on the circular birefringence. In order to satisfy the principle of reciprocity, as anticipated at the end of the last section, the sign of the rotation angle δ is conserved in the reciprocal case, while it is changed in the non-reciprocal case. This is in agreement with the physical considerations at the end of Section 4, based on the different behavior of polar and axial vectors in the presence of a mirror. In fact, the reciprocal circular birefringence gives a change of the rotation sign in the reversed direction, when observed in the same reference frame before and

Figure 9



Effect of the mirror reflection on the Poincaré sphere. To maintain the right-hand set even in the returning beam, the y axis is inverted. In consequence, the Poincaré sphere turns around axis $H-V$ (axis S_1) by an angle π . This operation is also called *binary rotation*.

after the mirror. However, for describing the evolution of the polarization, according to [30], we reverse axes y and z of the reference frame after the mirror, as shown by Fig. 9. We can recognize that the handedness of the reciprocal rotation with respect to axis z , which follows the propagation direction, is not changed. Because the two reference frames before and after the mirror have the same handedness, we have that the sign of δ is the same in both the forward and backward directions for the reciprocal case, while is opposite in the two different directions for the nonreciprocal case. We can summarize the effect of the mirror on the different kinds of birefringence by means of the following expressions:

$$R_{\rightarrow}^{\text{rec}}[\gamma, \phi] \xRightarrow{\text{inversion of propagation}} R_{\leftarrow}^{\text{rec}}[-\gamma, \phi], \quad (42)$$

$$R_{\rightarrow}^{\text{non-rec}}[\gamma, \phi] \xRightarrow{\text{inversion of propagation}} R_{\leftarrow}^{\text{non-rec}}[-\gamma, -\phi], \quad (43)$$

$$C_{\rightarrow}^{\text{rec}}[\delta] \xRightarrow{\text{inversion of propagation}} C_{\leftarrow}^{\text{rec}}[\delta], \quad (44)$$

$$C_{\rightarrow}^{\text{non-rec}}[\delta] \xRightarrow{\text{inversion of propagation}} C_{\leftarrow}^{\text{non-rec}}[-\delta]. \quad (45)$$

Exploiting these rules of transformation of the polarization operators, combined with the representation of the mirror action presented above, we can now study the evolution of the state of polarization in a generic retracing path composed by a series of birefringent plates and a common mirror. Such a study can be obviously carried out using both Jones matrices and Poincaré sphere representation.

For instance, considering the Jones matrix of the mirror given by Eq. (41), we can express the relation between the Jones vector J_{out} at the output of the retracing path and the Jones vector J_{in} at the input as

$$\begin{aligned} J_{\text{out}} &= M_{\leftarrow}^{\text{birefringence}} \cdot M_{\text{mirror}} \cdot M_{\rightarrow}^{\text{birefringence}} \cdot J_{\text{in}} \\ &= M_{\leftarrow}^{\text{birefringence}} \cdot \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot M_{\rightarrow}^{\text{birefringence}} \cdot J_{\text{in}}, \end{aligned} \quad (46)$$

where we have introduced the Jones matrices of the overall birefringence for both the forward and backward directions, which are related by the rules of Eqs. (42)–(45). We can see that, in general, the output state of polarization depends not only on the input state, but also on the birefringence of the circuit. In several applications the birefringence is randomly time dependent, and this dependence affects the output state of polarization, even if the input state of polarization is fixed, according to the expression

$$J_{\text{out}}(t) = M_{\leftarrow}^{\text{birefringence}}(t) \cdot \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot M_{\rightarrow}^{\text{birefringence}}(t) \cdot J_{\text{in}}. \quad (47)$$

This is the cause of many drawbacks in optical circuits requiring a stable polarization. In the next section we will investigate some schemes of birefringence compensation, allowing for the cancellation of the birefringence effect and for the recovery of a stable output state of polarization.

6. BINARY ROTATIONS AND GENERALIZED MIRRORS

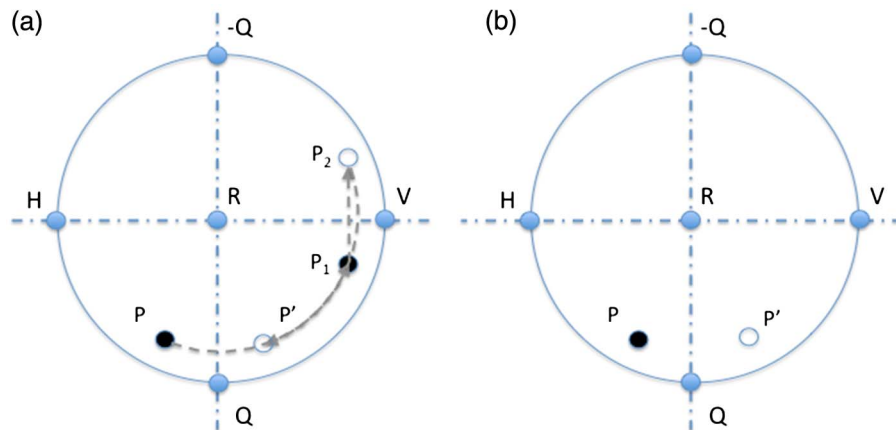
Before we proceed, let us consider in deeper detail the mirror operation. We have seen in the previous section that the action of the mirror can be represented on the Poincaré sphere by a rotation by an angle π (also named, by Altmann, *binary rotation* [12]) around the axis S_1 , as shown in Fig. 9. Such a binary rotation leaves unchanged the Stokes parameter S_1 , while it reverses the sign of the other two parameters S_2 and S_3 . The binary rotation offers the important advantage that is “passive;” in other words, it does not require any control or feedback for setting the exit point as a function of the entrance one. This guarantees that if we use this operation for compensating for the birefringence, this is passive and “transparent” (thus, it occurs without need of measurement of the entrance state of polarization and it is not influenced by the other characteristics of the light beam).

Is it possible to extend the properties of the common mirror to the other two axes of the Poincaré sphere? In other words, since there is no reason to privilege the S_1 axis, we expect there could exist also other devices able to overturn the Poincaré sphere around axes S_2 and S_3 , while inverting the propagation direction. The answer is positive, and we can give the definition of *generalized mirror* for an optical device by inverting the propagation direction and representing on the Poincaré sphere by a binary rotation around a generic axis passing through the center of the sphere.

The generalized mirror represented by a binary rotation on the Poincaré sphere around the S_2 axis is realized by combining a Faraday rotator, that is, a device realized by a material showing a magneto-optical activity as at point f) in Section 4, giving a polarization rotation $\delta = 45^\circ$, together with a common mirror. The related compensation properties were proposed by one of the authors in 1989 [32]. This paper drew for the first time the attention on the generalized compensation properties of the device, showed its representation on the Poincaré sphere, and recovered its matrix expression. Actually, two years before a similar device, exploited to compensate for the birefringence fluctuations introduced by an optical fiber, was independently described by Novikov’s group [33,34]. Furthermore, in the literature was reported the use of the Faraday rotator since 1980 by Giuliani and Ristori [35], in order to compensate for the birefringence originated in a laser cavity.

In the following we name the above-described generalized mirror a mirrored Faraday rotator (MFR) (in the literature, it is also known as Faraday rotator mirror). In Fig. 10

Figure 10



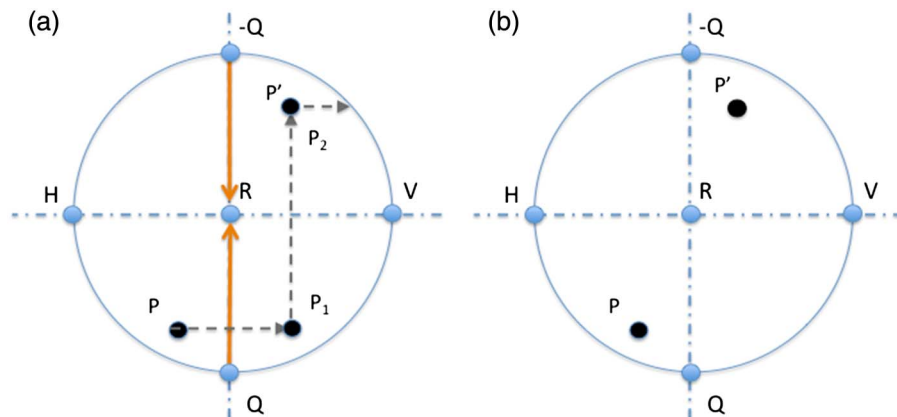
(a) Original state of polarization P ; (b) reflected state of polarization P' by a MFR (the generalized mirror represented by the binary rotation around axis S_2). The evolution on the Poincaré sphere of the state of polarization is highlighted by the dashed line.

it is possible to follow on the Poincaré sphere the action of the MFR. In this polar picture we use the full dots for the points in the foreground on the north hemisphere (right elliptical states) and the empty dots for the points in the background on the south hemisphere (left elliptical states). The Faraday rotator gives a counter-clockwise rotation around the S_3 axis by an angle $2\delta = 90^\circ$, moving the input point P in P_1 . The common mirror carries out a binary rotation around the S_1 axis, moving point P_1 into point P_2 . The beam then retraces the Faraday rotator, which is nonreciprocal, and according to Eq. (41) it gives a clockwise rotation around the S_3 axis by an angle $2\delta = 90^\circ$, producing the final point P' . From Fig. 10(b) we can see that the action of the MFR is to change the sign of Stokes parameters S_1 and S_3 , while maintaining constant the Stokes parameter S_2 , actually corresponding to a binary rotation around the S_2 axis. Moreover, it is well known that MFR possesses the very remarkable property of compensating for any reciprocal birefringence, both linear and circular.

Another generalized mirror, which produces a binary rotation on the Poincaré sphere around the S_3 axis, has been recently introduced. It is realized [11] by combining a quarter-wave plate with diagonal birefringence axes to a common mirror and can be called a mirrored quarter-wave plate (MQW). In Fig. 11 it is possible to follow on the Poincaré sphere the action of the MQW. The quarter-wave plate gives a counter-clockwise rotation around the S_2 axis by 90° , moving the input point P_{in} in P_1 . The common mirror carries out a binary rotation around the S_1 axis, moving point P_1 into point P_2 . The beam then retraces the quarter-wave plate, which is reciprocal, but inverts the azimuth of the birefringence vector in the mirrored reference frame, according to Eq. (38), hence giving a clockwise rotation around the S_2 axis by 90° , producing the final point P_{out} . It is evident from Fig. 11(b) that the action of the MQW is to change the sign of the Stokes parameters S_1 and S_2 , while maintaining constant the Stokes parameter S_3 , corresponding to a binary rotation around the S_3 axis. In [11] it is shown that MQW compensates for the nonreciprocal circular birefringence. Another possible way of realizing this kind of generalized mirror is to consider a Porro prism with an edge oriented at 45° [36].

After the introduction of generalized mirrors, let us return to the problem of the birefringence compensation in a retracing path. In general, since any birefringence is represented by a rotation vector on the Poincaré sphere, the criterion of self-compensation is to ensure that the birefringence rotation acting in the backward

Figure 11



(a) Original state of polarization P ; (b) reflected state of polarization P' by a MQW (the generalized mirror represented by the binary rotation around axis S_3). The evolution on the Poincaré sphere of the state of polarization is evidenced by the dashed line.

direction can be equivalently considered *before* the generalized mirror as the inverse of the birefringence rotation acting in the forward direction. In mathematical terms, we have a self-compensation if Eq. (47) becomes

$$\begin{aligned}
 J_{\text{out}} &= M_{\text{birefringence} \leftarrow}(t) \cdot M_{\text{generalized mirror}} \cdot M_{\text{birefringence} \rightarrow}(t) \cdot J_{\text{in}} \\
 &= M_{\text{generalized mirror}} \cdot M_{\text{birefringence} \rightarrow}^{-1}(t) \cdot M_{\text{birefringence} \rightarrow}(t) \cdot J_{\text{in}} \\
 &= M_{\text{generalized mirror}} \cdot J_{\text{in}}.
 \end{aligned} \tag{48}$$

In a more compact operatorial form, this self-compensation criterion can be written as

$$M_{\text{birefringence} \leftarrow}(t) \cdot M_{\text{generalized mirror}} = M_{\text{generalized mirror}} \cdot M_{\text{birefringence} \rightarrow}(t). \tag{49}$$

This strategy presents some similarity with the use of the phase-conjugated mirror (PCM) for cancelling the phase perturbations that occur in retracing optical circuits [37–41], because even in that case, the cancellation (here equivalent to the compensation) is obtained by a self-tracing.

The criterion expressed by Eqs. (48) and (49) is satisfied in two geometrical conditions on the Poincaré sphere, leading to the two following *rules of compensation*:

- **Rule 1.** If the birefringence rotation in the backward direction is the inverse of the birefringence rotation in the forward direction, then the compensating mirror is the generalized mirror whose binary rotation occurs around the same axis of the birefringence rotation.
- **Rule 2.** If the birefringence rotation in the backward direction is equal to the birefringence rotation in the forward direction, then the compensating mirror are the generalized mirrors whose binary rotation occurs around the axis orthogonal to the birefringence rotation.

Thanks to the isotropy of the Poincaré sphere (that is the equivalence between all the directions), we can demonstrate these two rules of compensation for the nonreciprocal and reciprocal circular birefringence, respectively, without loss of generality. Therefore, we consider in the following such two particular cases.

- A nonreciprocal circular birefringence is represented on the Poincaré sphere (see Fig. 12) by a rotation around S_3 with opposite angles in forward and backward propagation, as described by the operators $C[\delta]$ and $C[-\delta]$, respectively. Introducing the operator symbol Π_3 for the generalized mirror MQW corresponding to the binary rotation around the axis S_3 , we have

$$C[-\delta] \cdot \Pi_3 \cdot C[\delta] = \Pi_3 \cdot C[-\delta] \cdot C[\delta] = \Pi_3, \tag{50}$$

which demonstrates the birefringence compensation, owing to the commutation property of the product between two rotation operators around the same axis. Indeed, such a property makes the criteria [Eqs. (48) and (49)] satisfied in the case of nonreciprocal circular birefringence.

- A reciprocal circular birefringence is represented on the Poincaré sphere (see Fig. 13) by a rotation around S_3 with opposite angles in forward and backward propagation, described by the operators $C[\delta]$ and $C[-\delta]$, respectively. Introducing the operator symbol Π_n for the generalized mirrors corresponding to the binary rotation around the axis S_n , we have

$$\underline{C}[\delta] \cdot \Pi_n \cdot \underline{C}[\delta] = \Pi_n \cdot \underline{C}[-\delta] \cdot \underline{C}[\delta] = \Pi_n \quad (\text{for } n = 1, 2), \quad (51)$$

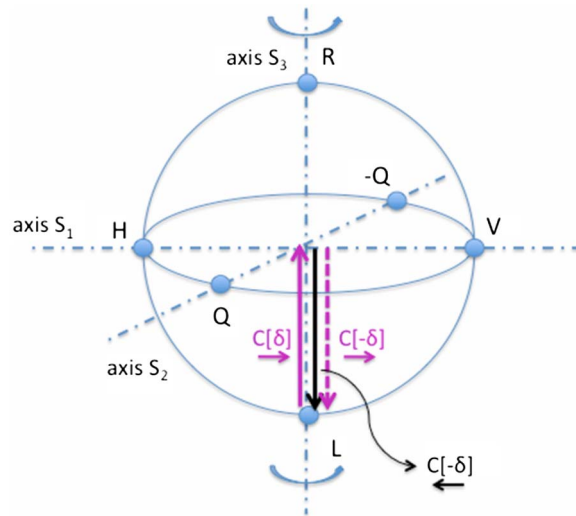
which demonstrates the birefringence compensation by a binary rotation around an axis orthogonal to the axis of birefringence rotation. In Eq. (51) is exploited the property that the product between a binary rotation and a rotation by a generic angle δ , in the case of orthogonal axes of rotations, is equal to the product in inverse order, provided that δ is replaced by the opposite angle $-\delta$, as can be seen by a simple geometrical demonstration on the Poincaré sphere. Indeed, such a property makes the criteria [Eqs. (48) and (49)] satisfied in case of reciprocal circular birefringence.

By applying the above two rules, we can identify what kinds of birefringence (linear or circular, reciprocal or nonreciprocal) are compensated for by the generalized mirrors represented by the operators Π_1 , Π_2 , and Π_3 (i.e., the binary rotations around the three reference axes in the abstract space of the Stokes parameters), as reported in Table 1.

Actually, this table refers to the “rectilinear” and “diagonal” cases of linear birefringence, represented by rotations around axes S_1 and S_2 , respectively. Nevertheless, it is possible to formulate the following further rule of compensation, which applies to a generic linear birefringence, represented by a rotation around an axis on the equatorial plane of the Poincaré sphere and passing through the center.

- **Rule 3.** If a linear birefringence, represented by a given rotation axis, is compensated for by a binary rotation around an axis on the equatorial plane (like, for instance, a binary rotation around S_1 or S_2), then any linear birefringence is compensated for, whatever is the rotation axis on the equatorial plane. The reason is that the change of the orientation of the rotation axis corresponding to a linear birefringence is obtained, according to Eq. (38), by applying operators equivalent

Figure 12



Representation on the Poincaré sphere of the compensation of the nonreciprocal circular birefringence by the generalized mirror corresponding to a binary rotation around axis S_3 . The forward birefringence is described by the continuous purple rotation vector, which is opposite to the dashed purple vector obtained by equivalently moving the backward circular birefringence (continuous black vector) before the mirror.

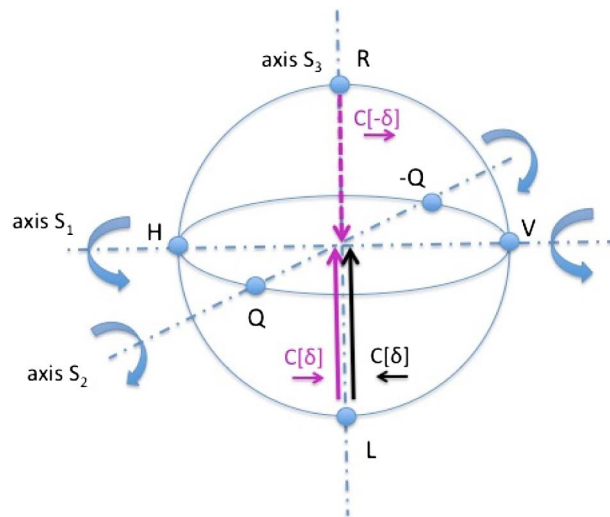
to a reciprocal circular birefringence, which is actually compensated by a binary rotation around an axis on the equatorial plane, owing to Rule 2.

From the results of Table 1, taking into account also Rule 3, we may formulate some observations.

- The rotation vectors of the linear birefringence show a different behavior for propagation inversion along the two different axes S_1 and S_2 . In fact, this anomaly is related to the inversion of the reference axis y , and not of the reference axis x , by effect of the propagation inversion.
- The reciprocal linear birefringence of any orientation, corresponding to a generic rotation vector on the equatorial plane (including the two particular cases evidenced by Table 1 of rotation around the axes S_1 and S_2) is compensated for by the same device, the MFR. Moreover, since it compensates even for the circular birefringence, the MFR can be considered a universal compensating generalized mirror for the reciprocal birefringence.
- As far as concern the nonreciprocal birefringence, we have not a symmetric situation. In fact the nonreciprocal linear birefringence of any orientation is compensated for by the mirror M, but the same mirror does not compensate for the nonreciprocal circular birefringence. The latter is compensated for by the MQW generalized mirror.

In the following section, we will explore more in detail the compensating action of the three above-generalized mirrors, represented by the binary rotations Π_1 , Π_2 , and Π_3 along the three axes of the Stokes space. We remark that such binary rotation offers the important advantage that they are realized in a passive way; in other words, they do not require any control or feedback for recovering the changes of the exit state of polarization. This guarantees that the compensation of the birefringence occurs for any input

Figure 13



Representation on the Poincaré sphere of the compensation of the reciprocal circular birefringence by the generalized mirror corresponding to a binary rotation around either axis S_1 or axis S_2 . The forward birefringence is described by the continuous purple rotation vector, which is opposite to the dashed purple vector obtained by equivalently moving the backward circular birefringence (continuous black vector) before the mirror.

Table 1. Compensation Strategies for Linear and Circular Birefringence^{ab}

Birefringence	Birefringence Rotation Axis on the Poincaré Sphere	Birefringence Vector Under Propagation Inversion	Birefringence Vector Sign Change Under Propagation Inversion	Binary Rotation Axis of the Compensating Generalized Mirror	Generalized Mirror Compensating Devices	
Linear	S_1	Reciprocal	no	S_2 or S_3	MFR	MQW
		Non-reciprocal	yes	S_1	M	
	S_2	Reciprocal	yes	S_2	MFR	
		Non-reciprocal	no	S_1 or S_3	M	MQW
Circular	S_3	Reciprocal	no	S_1 or S_2	M	MFR
		Non-reciprocal	yes	S_3	MQW	

^aLinear and circular birefringence can be represented by rotations around the three main axes of the Poincaré sphere.

^bIn the inverse propagation, the birefringence vectors either can or cannot change sign (depending on their reciprocity character). This fact causes different ways of compensation by a binary rotation and, as a consequence, different ways to obtain the compensation of the birefringence by the simple use of generalized mirrors.

state and is transparent (that is, not influenced by the other characteristics of the light beam).

7. EXAMPLES OF BIREFRINGENCE COMPENSATION IN RETRACING CIRCUITS

Let us examine in detail the action of the three generalized mirrors introduced in the previous section and interpreted as binary rotations on the Poincaré sphere. We can note the very interesting property that the Jones matrices corresponding to the binary rotation operators Π_1 , Π_2 , and Π_3 are the Pauli matrices σ_1 , σ_2 , and σ_3 , respectively. In the following, we will analyze the different cases of birefringence compensation in retracing circuits by also using the Jones formalism, in addition to the polar view of the Poincaré sphere. We will also review the experimental evidence of the predicted effects.

7.1. Common Mirror

Let us begin from the common mirror, which we have seen to be represented on the Poincaré sphere by a binary rotation around the axis S_1 . The Jones matrix of the mirror is shown by Eq. (41) and we can note that is identical to the Pauli matrix σ_1 . We have, in fact,

$$M_{\text{mirror}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \sigma_1. \quad (52)$$

According to the general rules of compensation seen in the last section, the common mirror compensates for both the reciprocal circular birefringence and the nonreciprocal linear birefringence.

In the case of reciprocal circular birefringence, that is, the case d) in Section 4, giving a polarization azimuth rotation by an angle δ , we can write the matrix of the retracing optical circuit as

$$M_{\text{retracing}} = C_{\leftarrow}^{\text{rec}}[\delta] \cdot \sigma_1 \cdot C_{\rightarrow}^{\text{rec}}[\delta] = C[\delta] \cdot \sigma_1 \cdot C[\delta] = \sigma_1, \quad (53)$$

because it is easy to verify that holds the matrix identity

$$C[\delta] \cdot \sigma_1 \cdot C[\delta] = \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \sigma_1. \quad (54)$$

Hence Eq. (53) states the effective self-compensation of the reciprocal circular birefringence in the retracing optical circuit operated by the common mirror. This property can also be seen in the Poincaré sphere representation through the polar view sketched in Fig. 14.

Examples of reciprocal circular birefringence are very common. This type of circular birefringence was first discovered by Arago in quartz in 1811 and described by Biot in 1812 [22]. It is also known as optical activity, as seen in Subsection 4.2, and is present in many crystals, as well as in organic molecules and liquid crystals. It can also be induced by an external agent, the most common being a torsional stress applied to a homogeneous and isotropic material, as a glass. Also, the fused silica of the optical fiber becomes optically active when twisted. Ulrich demonstrated that the polarization rotation angle δ is proportional to the applied torsion by the relation

$$\delta = g\tau, \quad (55)$$

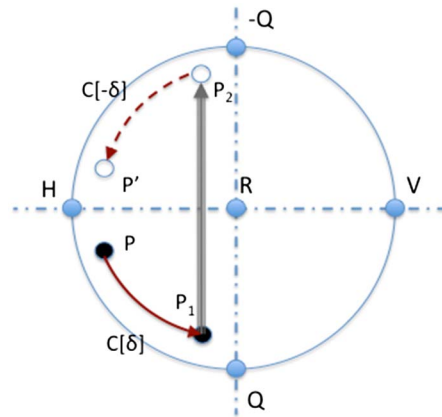
where τ is the angle of twist and g is the gyration parameter, a function of the elasto-optic coefficients p_{ij} of the material. For an amorphous material (like fused silica), it results that

$$g = -\frac{n^2}{2}(p_{11} - p_{12}) = -n^2 p_{44}, \quad (56)$$

and for a common single-mode fiber the parameter g is about 0.16.

In case of a nonreciprocal linear birefringence, that is, case d) in Section 4, with retardation ϕ and orientation θ , we can write the matrix of the retracing optical circuit as

Figure 14



Compensation for the reciprocal circular birefringence by a common mirror. The rail vector represents the mirror action.

nonreciprocal linear birefringence by a common mirror has not yet been experimentally proved.

7.2. Mirrored Faraday Rotator

The second generalized mirror that we analyze in detail is the MFR, which we have seen to be represented on the Poincaré sphere by a binary rotation around the axis S_2 . According to the general rules of compensation seen in the last section, this generalized mirror compensates for both the reciprocal circular birefringence and the reciprocal linear birefringence. According to [32], we can write the Jones matrix of the MFR, apart from a not relevant multiplicative phase factor, as the anti-diagonal matrix

$$M_{\text{MFR}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \sigma_2, \quad (61)$$

which is even identical to the Pauli matrix σ_2 . This matrix expresses the exchange between the horizontal and vertical states of polarization after the reversal of direction operated by the MFR.

In the case of reciprocal circular birefringence, that is, the cases d) and e) in Section 4, the matrix of the retracing optical circuit including the MFR is

$$M_{\text{retracing}} = \overset{\leftarrow}{C}[\delta]^{\text{rec}} \cdot \sigma_2 \cdot \vec{C}[\delta]^{\text{rec}} = C[\delta] \cdot \sigma_2 \cdot C[\delta] = \sigma_2, \quad (62)$$

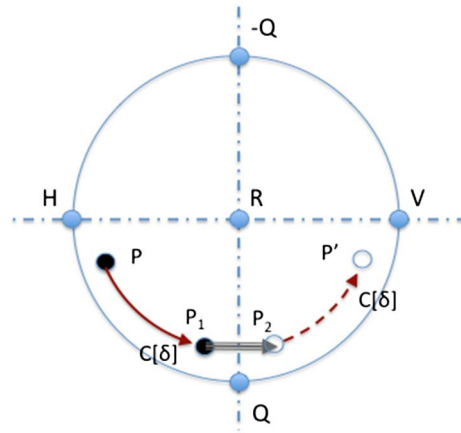
showing the compensation, thanks to the matrix identity

$$C[\delta] \cdot \sigma_2 \cdot C[\delta] = \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \sigma_2. \quad (63)$$

The compensation effect is also shown in Fig. 16.

Moreover, considering the effect of the MFR on reciprocal linear birefringence, that is, the cases a) and b) in Section 4, we have

Figure 16



Compensation for the reciprocal circular birefringence by the MFR generalized mirror. The rail vector represents the MFR action. If point P coincides with the horizontal polarization, the light returns vertically polarized, with no trace of whatever birefringence.

7.3. Mirrored Quarter-Wave Plate

Finally, we consider the third generalized mirror, given by the MQW, which is represented on the Poincaré sphere by a binary rotation around axis S_3 . The Jones matrix of the MQW, according to [11], apart from a not relevant multiplicative phase factor, can be considered identical to the Pauli matrix σ_3 , that is,

$$M_{\text{MQW}} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \sigma_3. \quad (66)$$

We know from Table 1, according to the general rules of compensation in terms of binary rotations seen in the last section, that the MQW generalized mirror compensates for the nonreciprocal circular birefringence.

The Jones matrix for the optical circuit including the MQW and retracing a nonreciprocal circular birefringence is

$$M_{\text{retracing}} = C_{\leftarrow}^{\text{non-rec}}[-\delta] \cdot \sigma_3 \cdot C_{\rightarrow}^{\text{non-rec}}[\delta] = C[-\delta] \cdot \sigma_3 \cdot C[\delta] = \sigma_3, \quad (67)$$

because we can write the Pauli matrix σ_3 as the product between the imaginary unit and the rotation matrix by an angle, that is,

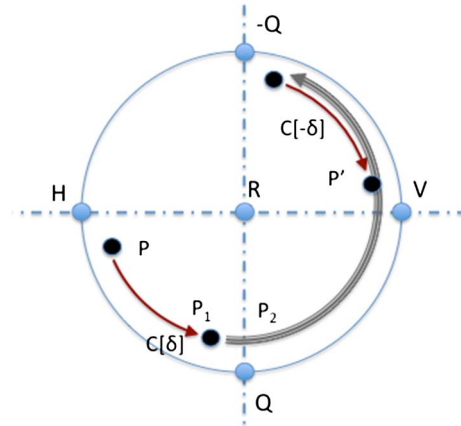
$$\sigma_3 = i \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = iC[\pi/2]. \quad (68)$$

Hence, Eq. (67) holds for the commutation of the rotator Jones matrices, being

$$C[-\delta] \cdot \sigma_3 \cdot C[\delta] = C[-\delta] \cdot C[\delta] \cdot \sigma_3 = \sigma_3. \quad (69)$$

Equation (67) shows the birefringence compensation of the nonreciprocal circular birefringence. Such a compensation was verified in [11] by a very recent experiment, showing the cancellation of the effect of a variable Faraday rotator (see also Fig. 18). The generalized mirror MQW is also equivalent to a Porro prism with its edge oriented

Figure 18



Compensation for the nonreciprocal circular birefringence by the MQW generalized mirror. The rail vector represents the MQW action. If point P coincides with the horizontal polarization, the light returns vertically polarized, with no trace of whatever birefringence.

at an angle of $\pi/4$, whose compensation property of nonreciprocal circular birefringence was experimented with in [36].

Moreover, following Table 1, the MQW compensates also for a reciprocal linear birefringence with fast axis azimuth $\gamma = 0$ (thus represented on the Poincaré sphere by a rotation around axis S_1) and for a nonreciprocal linear birefringence with fast axis azimuth $\gamma = \pi/4$ (thus represented on the Poincaré sphere by a rotation around axis S_2). The Jones matrices of the retracing circuit in the former case are in fact

$$M_{\text{retracing}} = R_{\leftarrow}^{\text{rec}}[0, \phi] \cdot \sigma_3 \cdot R_{\rightarrow}^{\text{rec}}[0, \phi] = \sigma_3. \quad (70)$$

For the latter case, we can write

$$\begin{aligned} M_{\text{retracing}} &= R_{\leftarrow}^{\text{non-rec}}[-\pi/4, -\phi] \cdot \sigma_3 \cdot R_{\rightarrow}^{\text{non-rec}}[\pi/4, \phi] \\ &= C[-\pi/4] \cdot R[0, -\phi] \cdot C[\pi/4] \cdot \sigma_3 \cdot C[\pi/4] \cdot R[0, \phi] \cdot C[-\pi/4]. \end{aligned} \quad (71)$$

Taking into account Eq. (68) we have

$$C[\pi/4] \cdot \sigma_3 \cdot C[\pi/4] = C[\pi/4] \cdot iC[\pi/2] \cdot C[\pi/4] = iC[\pi] = -iI, \quad (72)$$

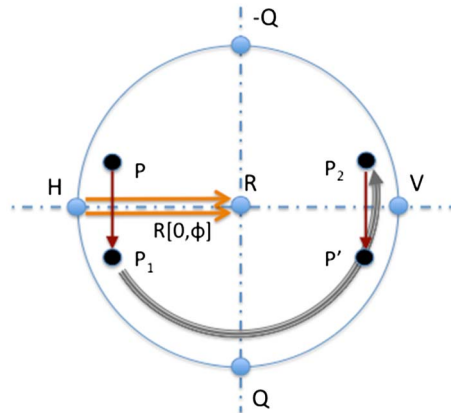
where I is the identity matrix. By substituting Eq. (72) into Eq. (71) we obtain

$$M_{\text{retracing}} = -iC[-\pi/4] \cdot R[0, -\phi] \cdot R[0, \phi] \cdot C[-\pi/4] = -iC[-\pi/2] = \sigma_3. \quad (73)$$

Hence Eqs. (70) and (73) demonstrate the birefringence compensation in the two respective cases. Figures 19 and 20 illustrate these compensations.

Table 2 summarizes the main type of birefringence as introduced in Section 4 and the relative compensation actuated by the generalized mirror. Some cancellation operations are still to be experimentally investigated, to the best of our knowledge.

Figure 19

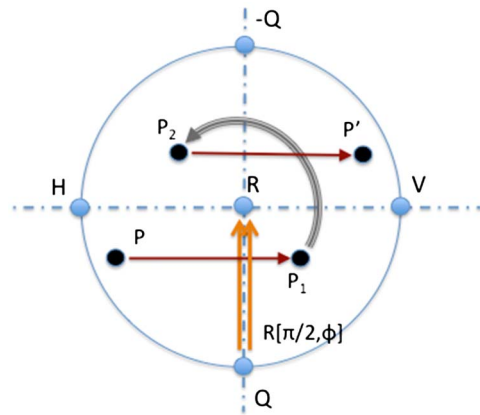


Compensation for the reciprocal linear birefringence $R[0, \phi]$ by the MQW generalized mirror. The rail vector represents the MQW action. If point P coincides with the horizontal polarization, the light returns vertically polarized, with no trace of whatever birefringence.

8. CONCLUSIONS

In Section 1 we pointed out that the aim of the paper is to identify, if they exist, some optical circuits that can remove the birefringence impact on the polarization of propagating beams. Through the paper we discover that, in order to obtain this objective in a passive mode (that is, without the active control of the state of the polarization at the input or at the output), it is necessary to use a *self-compensation* approach. But this was only a necessary condition. It is not sufficient because the regular mirror does not hold all the symmetry properties necessary for a full compensation of all the kinds of birefringence. In fact, as represented in Table 1, linear and circular birefringence induce rotations around the three principal axes of the Poincaré sphere. Hence, in order

Figure 20



Compensation for the reciprocal linear birefringence $R[\pi/2, \phi]$ by the MQW generalized mirror. The rail vector represents the MQW action.

Table 2. Birefringence in the Propagation Space^{ab}

	Mechanical agent		Electric agent		Magnetic agent	
Origin	Twist in amorphous and cubic crystals	Bending in amorphous and cubic crystals	Electrogyration	Pockels effect	Faraday effect	Non-reciprocal birefringence
Example of material	Glass	Glass	Quartz	KDP	Glass and quartz	LiIO ₃ and KDP
Type of birefringence	Circular birefringence proportional to M	Linear birefringence proportional to M	Circular birefringence proportional to E	Linear birefringence proportional to E	Circular birefringence proportional to H	Linear birefringence proportional to H
Poincaré sphere representation	Vector aligned to the polar axis	Vector lying on the equatorial plane	Vector aligned to the polar axis	Vector lying on the equatorial plane	Vector aligned to the polar axis	Vector lying on the equatorial plane
Propagation characteristic	Reciprocal	Reciprocal	Reciprocal	Reciprocal	Non-reciprocal	Non-reciprocal
Cancelled by	MFR and M	MFR	MFR*	MFR*	MQW	M*
Quoted in the text	d)	a)	e)	b)	f)	c)

^aIn the table, M means moment of an applied force, E means electrical field, and H means magnetic field.

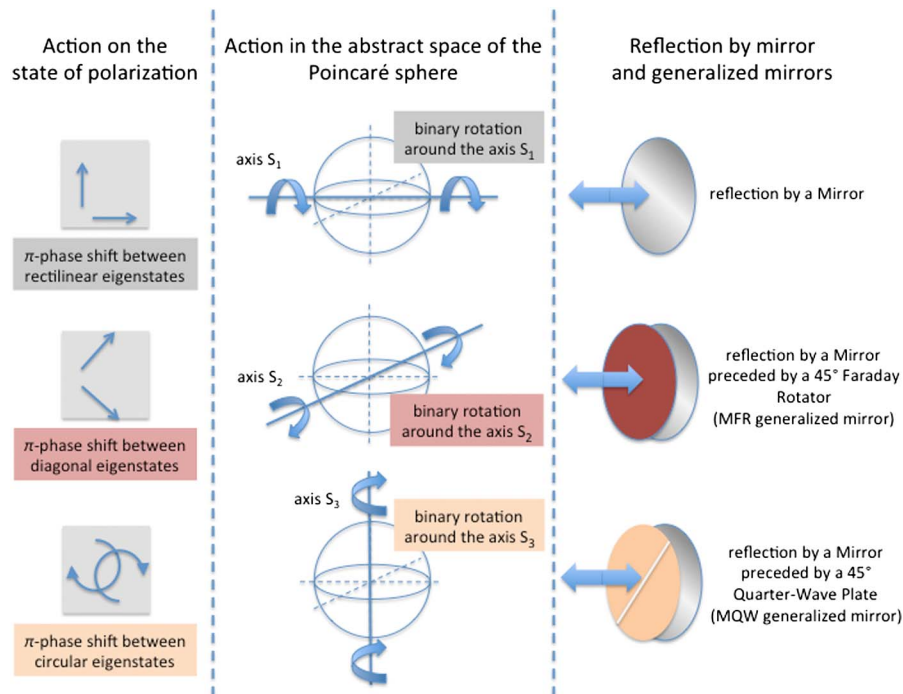
^bIn principle, by using the proper generalized mirrors, all the main types of natural and induced birefringence (both in the free- and guided-propagation space) can be compensated for. The asterisk points out the absence of literature references about the specific compensation, to the best of our knowledge.

to realize a self-compensation, it is necessary to introduce two further generalized mirrors (the MFR and MQW), which complete the triad of mirrors that permit a binary rotation around the three main axes of the Poincaré sphere. By means of the above rotations and by the properties of the retracing circuits it is possible to compensate for *all* the kinds of birefringence, both reciprocal and nonreciprocal. Of course, there does not exist a unique mirror that can do this, but we need specialized solutions (that is, the use of a specific mirror) for each specific condition of birefringence. Table 2 summarizes the results of the paper and points out that all kinds of birefringence occurring in practical optical circuits are compensated got by the suitable use of a mirror triad in a retracing circuit configuration. The table also points out (asterisks) that there is no evidence in the literature, to the best of our knowledge, of certain types of compensation (indeed, apart from the Pockels' effect, they refer to rather uncommon cases of birefringence).

The action of the three mirrors can be better understood in the abstract space of the Poincaré sphere (see Fig. 21). The common mirror makes a binary rotation (that is, a rotation by an angle π) around axis S_1 of the Poincaré sphere and leaves invariant the rectilinear (i.e., horizontal and vertical) states of polarization. We have demonstrated that the action of the mirror is only one of the three possible; the others are given by the MFR and MQW, which act on the other two axes, S_2 and S_3 , of the sphere (and leave invariant the diagonal/anti-diagonal and right/left circular states of polarization, respectively). So, in the abstract space of the Poincaré sphere representation, the action of the three mirrors can be seen in a unitary frame.

In summary, we have seen that the combination of “polarization, mirror, and reciprocity” generates a complex system of symmetries not obvious to understand. The representation of the Poincaré sphere once again proves to be the most effective tool to

Figure 21



In the abstract space of the Poincaré sphere, the action of the three mirrors (the common mirror and the two generalized mirrors MFR and MQW) appears equivalent. The figure shows also the effects of the mirror triad on the three couples of the polarization eigenstates (in the real space).

solve this complexity because it is the appropriate space for the representation of the spin of the photon.

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REFERENCES AND NOTES

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are only the two values ± 1 , showing an exception to the previous rule. In actuality, the photon is a massless particle always moving at the velocity of the light, and owing to this constraint (called the *transversality constraint*) the value 0 is not a permitted spin eigenvalue for the photon; see [20] as an example.

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